

A Secure Takagi–Sugeno Fuzzy Controller with Dual Lyapunov–LMI Synthesis for Cyber–Physical Systems under Cyberattacks

Abstract

In Cyber Physical Systems (CPS), such as autonomous vehicles, industrial robotics, and smart grids, cyber threats such as False Data Injection (FDI), Denial-of-Service (DoS), and Distributed DoS (DDoS) attacks have become increasingly regular. By altering with control signals or distorting sensor feedback, these hostile intrusions may result in instability or risky operation. Traditional fuzzy and adaptive controllers are vulnerable to attacks because they depend on reliable communication.

In this proposal we propose a Secure Takagi Sugeno Fuzzy Controller (STSFC) that integrates cybersecurity awareness into the fuzzy inference process. The controller consists of three key components: a fuzzy observer for anomaly detection, a security-weighted inference mechanism that adjusts rule activations based on real-time trust evaluation, and a dual-surface Lyapunov LMI framework that guarantees stability under both normal and attacked conditions.

The proposed STSFC is resistant to adversarial manipulation and data poisoning since it is model-based and non-adaptive. Applying Lyapunov LMI conditions, stability and input-to-state stability will be analytically demonstrated.

The STSFC will be tested and implemented on an inverted pendulum system, a benchmark nonlinear platform intended to verify control robustness under cyberattacks and external disturbances, in order to assess its performance. A universal, self-defensive fuzzy control framework that improves the resilience and dependability of networked control systems is anticipated, along with sustained stability and reduced detection error.

Keywords: Takagi–Sugeno fuzzy control, nonlinear system stabilization, resilient control, cyber–physical systems, Denial-of-Service attack.

1. Introduction

1.1 Background

Cyber-Physical Systems (CPS), which integrate communication networks, computational intelligence, and physical processes into a single interactive framework, have become essential to modern infrastructure. They can be found in a variety of applications, including intelligent transportation systems, smart networks, industrial robotic manipulators, and autonomous aerial vehicles, [1], [2], [3].

The feedback control loop, which continuously exchanges sensor readings, control signals, and actuation orders via digital communication channels, is the core idea behind CPS.

A CPS's general dynamic model can be written as follows.

$$\begin{aligned}\dot{x}(t) &= f(x(t), u(t), t), \\ y(t) &= g(x(t), u(t), t),\end{aligned}$$

where $x(t) \in \mathbb{R}^n$ is the system state, $u(t) \in \mathbb{R}^m$ is the control input, and $y(t)$ denotes the measured output.

While the network's connectivity grows, CPS is increasingly vulnerable to cyberattacks that threaten synchronization of time and data integrity. Attacks like Denial-of-Service (DoS), Distributed DoS (DDoS), and False Data Injection (FDI) can skew sensor feedback or block control commands, causing the system to become unreliable or perform adversely [4], [5].

To mathematically model these attacks:

$$\hat{x}(t) = f(x(t), u(t), a(t)), y(t) = g(x(t), a(t)),$$

where $a(t)$ represents the cyberattack influence on communication channels. According to recent research, even small, invisible assaults can cause nonlinear CPS to become unstable if they go undetected [6], [7].

1.2 Takagi–Sugeno (TS) Fuzzy Control Framework

Takagi–Sugeno (T-S) fuzzy control has been successfully applied to nonlinear systems; [8] utilized a Sum-of-Squares (SOS) technique to guarantee performance and stability. In the same way, [9] implemented T-S fuzzy control for an overhead crane system, emphasizing assured cost management to maximize performance within realistic limitations.

The Takagi–Sugeno (TS) fuzzy control framework is a powerful tool for controlling nonlinear system dynamics via rule-based linearization. Instead of globally approximating a nonlinear function, it expresses the plant as a weighted sum of linear subsystems [2].

$$\dot{x}(t) = \sum_{i=1}^r h_i(z(t))(A_i x(t) + B_i u(t)),$$

$$\sum_{i=1}^r h_i(z(t)) = 1$$

where $h_i(z(t))$ are normalized membership functions determined by premise variables $z(t)$. Each fuzzy rule is written as:

Rule i : IF $z_1(t)$ is F_{11} ... $z_p(t)$ is F_{ip} THEN $\dot{x}(t) = A_i x(t) + B_i u(t)$.

The fuzzy control law is synthesized as:

$$u(t) = \sum_{i=1}^r h_i(z(t)) K_i x(t)$$

where K_i are feedback gains obtained by solving Linear Matrix Inequalities (LMIs):

$$(A_i + B_i K_i)^T P + P(A_i + B_i K_i) + Q < 0$$

According to [10], [11], this framework allows for the seamless mixing of several local linear models to capture nonlinear behavior.

However, accurate premise variables $z(t)$ are assumed by traditional TS fuzzy systems. According to [12], [13], cyberattacks have the potential to corrupt these variables, resulting in inappropriate rule activation and destabilization.

1.3 Vulnerabilities under Cyberattacks

The membership functions $h_i(z(t))$ get altered and incorrect fuzzy rules are activated when an attack corrupts $z(t)$ or sensor data $y(t)$. In this case, faking angular position or velocity data in an inverted pendulum could lead to faulty control actions, which would cause the pendulum to drop.

Typical attack representations include:

- FDI attack: $\hat{y}(t) = y(t) + \Gamma_y \alpha_y(t)$;
- DoS attack: $u(t) = \rho(t) u_c(t)$, $\rho(t) \in \{0, 1\}$;
- DDoS: network congestion delaying packets, $\hat{\rho}(t) = -\alpha \rho(t) + \beta \sum_{j=1}^{N_2} \sigma_j(t)$.

Such attacks have been reported in smart grids industrial control systems and autonomous robotics [14]. These weaknesses show that rather than relying on external intrusion detection or adaptive learning, a controller must incorporate security awareness directly into its fuzzy inference process. Conventional TS controllers malfunction when signals become corrupted since they rely on precise premise parameters.

1.4 Motivation

While existing fuzzy control techniques such as observer-based, adaptive, and event-triggered controllers improve efficiency and robustness, they do not incorporate cybersecurity measures into the fuzzy inference process. The results of attacks are not utilized to alter control choices in real time when they are discovered [15]. A secure, non-adaptive fuzzy controller that can both identify and mitigate threats within its inference process is therefore desperately needed.

The goal of this proposal is to create a Secure Takagi–Sugeno Fuzzy Controller (STSFC) that achieves self-defensive control stability by directly integrating trust and detection input into fuzzy rule weighting. Because of its great sensitivity to sensor corruption and control delays, the inverted pendulum system a typical nonlinear benchmark will be used for evaluation.

1.5 Objectives

The goal of this proposal is to develop a Secure Takagi–Sugeno Fuzzy Controller (STSFC) that can protect CPS from cyberattacks without affecting performance or stability.

The main objectives of this research are:

- To develop a fuzzy observer using residual estimation for anomaly detection.
- To implement a security-weighted inference technique that minimizes the impact of data corruption.
- To establish dual Lyapunov-LMI conditions that guarantee stability in both attack and normal modes.
- To deploy and evaluate the proposed controller on an inverted pendulum system that is vulnerable to DDoS, DoS, and FDI attacks.

The control architecture will be tested using an inverted pendulum, an established nonlinear benchmark that is very sensitive to sensor errors and delays.

2. Related Work

2.1 Observer-Based Secure Fuzzy Control

Observer-based fuzzy systems estimate unmeasured states and detect abnormal behavior. A common formulation is:

$$\dot{x}(t) = \sum_{i=1}^r h_i(z(t)) [A_i \hat{x}(t) + B_i u(t) + L_i (y(t) - C_i \hat{x}(t))]$$

with residuals $r_i(t) = y(t) - C_i \hat{f}(t)$.

The Collaborative Interaction Observer (CIO) was created by Huang et al. in 2024 to guarantee stability via

$$(A_i - L_i C_i)^T P + P(A_i - L_i C_i) + Q < 0$$

But the inference layer remains identical after a threat has been detected. While other designs, including adaptive observers [5], [6], [16], increase detection accuracy, they rely significantly on adaptive updates, which FDI may threaten. Therefore, current observers detect attacks but lack the ability to dynamically modify fuzzy control inference. An enhanced resilient observer for nonlinear systems was presented in [17], but it failed to update fuzzy rule activations during detection.

2.2 Event-Triggered and Adaptive Secure Fuzzy Control

Event-triggered control reduces data exchange under DoS conditions. The triggering condition is:

$$\|e(t)\|^2 > \delta \|x(t)\|^2$$

[15] proposed an adaptive event-triggered TS controller with LMIs ensuring:

$$\dot{V}(x) \leq -x^T Q x + \rho e^T e$$

Although efficient, it remains vulnerable to attack on adaptive parameters.

Despite their increase computational complexity, dynamic triggering systems [10] further reduce communication. Adaptive and neural fuzzy controllers [12], [13] rely on online learning.

$$\dot{\theta}(t) = -\gamma \frac{\partial J}{\partial \theta},$$

making them vulnerable to manipulation by attackers using tainted data streams.

Consequently, whereas event-triggered and adaptive approaches increase productivity, they are not intrinsically resistant to cyberattacks.

2.3 Resilient and Fault-Tolerant Fuzzy Control

Resilient fuzzy controllers maintain functionality despite faults:

$$\dot{x}(t) = \sum_{i=1}^r h_i(z(t)) [A_i x(t) + B_i (u(t) + \Delta u_i(t)) + E_i d(t)]$$

where $\Delta u_i(t)$ models' fault or attack effects. [11] ensured boundedness via primarily concentrating on adaptive attacks on networks, addressed identified vulnerabilities. consequently, robust approaches fail to incorporate detection input into fuzzy logic and preserve limited outputs

$$\dot{V}(x) \leq -x^T Q x + \eta \|\Delta t u(t)\|^2$$

Corrupted sources are still unchecked, regardless. [7] maintained stable fuzzy conclusion weights while improving detection accuracy.

2.4 Fuzzy–Lyapunov–LMI Stability

Standard Lyapunov–LMI stability for fuzzy systems is:

$$\dot{V}(x) = \sum_{i=1}^r h_i(z(t)) [x^T (A_i^T P + P A_i) x] < 0$$

The majority of studies, though, simply employ a single Lyapunov surface for static modes. Stability cannot be ensured under switching caused by cyberattacks [15]. According to [18], adding dual Lyapunov matrices improves resistance to time-varying DoS intervals.

The proposed STSFC introduces dual Lyapunov functions:

$$V_1(x) = x^T P_{1x}, V_2(x) = x^T P_{2x}$$

ensuring smooth switching stability between secure and compromised operation modes.

2.6 Research Implications

The above described literature that current TS fuzzy controllers lack security-integrated inference algorithms. In the face of active attacks, existing methods cannot change inference weights or decision rules, even though they can detect or tolerate errors.

Furthermore, no known method properly combines observer-based detection, dual Lyapunov-LMI synthesis, and trust-weighted fuzzy inference into a single architecture.

In order to close these gaps, this proposal develops a Secure Takagi–Sugeno Fuzzy Controller (STSFC) that:

- Provides mathematically guaranteed stability under switching through dual Lyapunov functions.

- Dynamically incorporates detection feedback into fuzzy rule weighting.
- Offers a computationally light, model-based, non-adaptive architecture appropriate for real-time CPS deployment.

3. Proposed Work

3.1 Overview

The proposed Secure Takagi–Sugeno Fuzzy Controller (STSFC) which is shown in figure 1, will provide an integrated architecture that combines stability assurance, defense, and attack detection into a single, integrated control framework. The STSFC actively assesses the reliability of incoming signals and dynamically modifies its control decisions based on real-time trust evaluation, whereas traditional fuzzy controllers presume trusted feedback.

The complete control structure will consist of three layers:

1. **Fuzzy Observer Layer:** will estimate true system states and detect anomalies through residual generation.
2. **Security-Weighted Fuzzy Inference Layer:** will reweight fuzzy rules using trust coefficients computed from observer residuals.
3. **Dual-Surface Lyapunov–LMI Stability Layer:** will guarantee global stability under both normal and attack-induced switching conditions.

The proposed approach will be implemented and tested on an **inverted pendulum** system as a representative Cyber–Physical System (CPS).

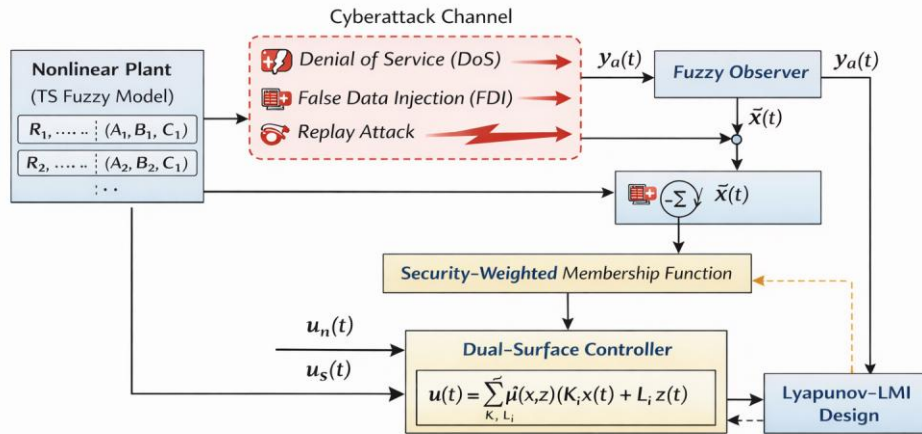


Fig. 1. Proposed Framework of TS-Fuzzy Controller

3.2 System Model under Cyberattacks

The nonlinear plant will be modeled using a Takagi–Sugeno fuzzy framework subject to potential cyberattacks:

$$\dot{x}(t) = \sum_{i=1}^r h_i(\bar{z}(t)) [A_i x(t) + B_i (u(t) + \Gamma_u a_u(t)) + E_i d(t)]$$

$$\bar{y}(t) = \sum_{i=1}^r h_i(\bar{z}(t)) (C_i x(t) + D_i a_y(t))$$

Where:

$x(t) \in \mathbb{R}^n$ is the system state vector,

$u(t) \in \mathbb{R}^m$ is the control input,

$a_u(t)$ and $a_y(t)$ denote attack signals injected into actuator and sensor channels,

$d(t)$ represents bounded external disturbances,

Γ_y is the actuator attack matrix, and $(\bar{z}(t))$ represents the compromised premise variable affected by sensor manipulation.

The attack-affected system dynamics will therefore differ from the nominal model due to the corruption of membership variables $h_i(\bar{z}(t))$.

3.3 Fuzzy Observer Design for Attack Detection

To detect compromised channels and reconstruct the true system state, a **fuzzy observer** will be designed as:

$$\dot{\hat{x}}(t) = \sum_{i=1}^r h_i(z(t)) [A_i \hat{x}(t) + B_i u(t) + L_i (\bar{y}(t) - C_i \hat{x}(t))],$$

$$r_i(t) = \bar{y}(t) - C_i \hat{x}(t),$$

where $r_i(t)$ represents the residual signal of each local model. Under normal operation, $r_i(t)$ will converge to zero, while large deviations will indicate an ongoing cyberattack.

The observer gains L_i will be derived from the following LMI conditions

$$(A_i - L_i C_i)^T P + P(A_i - L_i C_i) + Q < 0$$

This ensures asymptotic convergence of the estimation error $e(t) = x(t) - \hat{x}(t)$. A detection rule will be applied as:

$$\|r_i(t)\| > \varepsilon_i \Rightarrow \text{Attack detected on rule } i.$$

3.4 Security-Weighted Fuzzy Inference Mechanism

Once attacks are detected, their influence must be mitigated in the control process. To achieve this, the STSFC will employ **trust-weighted fuzzy inference**, where each fuzzy rule will be reweighted based on the observer's residual magnitude.

The trust level for the i^{th} rule will be computed as

$$\omega_i(t) = e^{-\lambda r_0(t)^2}, 0 < \omega_i(t) \leq 1,$$

where λ defines the sensitivity of the trust decay.

The modified (secure) membership functions will be expressed as:

$$\bar{h}_i(z(t)) = \frac{\omega_i(t) h_i(z(t))}{\sum_{j=1}^r \omega_j(t) h_j(z(t))}$$

As a result, rules corresponding to unreliable sensors or compromised data channels will automatically receive reduced influence.

The resulting secure fuzzy control law will be

$$u(t) = \sum_{i=1}^r K_i(z(t)) K_i \hat{x}(t),$$

where K_i are precomputed controller gains obtained from the LMI synthesis.

This mechanism enables the controller to autonomously adapt its inference to attack conditions without any adaptive learning, ensuring immunity to adversarial training or data poisoning.

3.5 Dual-Surface Lyapunov–LMI Stability Framework

Since cyberattacks can trigger instantaneous transitions between secure and attacked modes, the STSFC will utilize a dual Lyapunov framework.

Two Lyapunov functions will be defined as:

$$V_1(x) = x^T P_1 x, \quad (\text{secure mode})$$

$$V_2(x) = x^T P_2 x, \quad (\text{attack mode}).$$

The derivative of $V_j(x)$ under subsystem i will be

$$\dot{V}_j(x) = x^T [(A_i + B_i K_i)^T P_j + P_j (A_i + B_i K_i)] x + 2x^T P_j E_i d(t),$$

where $j \in \{1, 2\}$.

To ensure global stability, the following LMI must hold

$$\begin{bmatrix} (A_i + B_i K_i)^T P_j + P_j (A_i + B_i K_i) + Q_j & P_j E_i \\ E_i^T P_j & -\gamma I \end{bmatrix} < 0, j = 1, 2.$$

This condition guarantees:

$$\dot{V}_j(x) \leq -x^T Q_j x + \gamma \|d(t)\|^2.$$

Thus, Input-to-State Stability (ISS) will be preserved across both operational modes.

3.6 Input-to-State Stability under Attacks

The complete closed-loop system will satisfy the ISS condition:

$$\|x(t)\| \leq \beta(\|x(0)\|, t) + \sigma(\|a(t)\|),$$

Where β and σ are class - $K L$ and class- K functions, respectively.

This ensures that the system state remains bounded under persistent FDI or DDoS attacks.

3.8 Simulation and Validation on Inverted Pendulum System

To validate the proposed controller, simulations will be performed on an inverted pendulum a well-known benchmark for nonlinear control.

Its continuous-time dynamics are:

$$\begin{cases} \dot{x}_1 = x_2, \\ \dot{x}_2 = \frac{g}{l} \sin(x_1) + \frac{1}{ml^2} u(t), \end{cases}$$

where x_1 is the pendulum angle, x_2 the angular velocity, m the mass, l the length, and g gravitational acceleration.

The system will be linearized around multiple operating points to construct local models:

$$A_i = \begin{bmatrix} 0 & 1 \\ \cos(\theta_i) & 0 \end{bmatrix}, B_i = \begin{bmatrix} 0 \\ 1 \\ \frac{1}{mx^2} \end{bmatrix}.$$

4. Research Methodology

The purpose of this research will design, build, and verify a Secure Takagi–Sugeno Fuzzy Controller (STSFC) using a model-based, simulation-driven quantitative methodology.

The investigation will consist of simulation-based performance evaluation under various cyberattack scenarios, Lyapunov LMI stability analysis, and theoretical modeling. Throughout problem formulation to evaluation, the entire process is organized **into six logical steps** that guarantee accuracy, consistency, and reliable outcomes.

a) Problem Formulation and System Modeling

The inverted pendulum will be used as the test platform to model a nonlinear cyber-physical system at the start of the investigation. The dynamics of the pendulum will be linearized around multiple points of operation to produce a Takagi–Sugeno (TS) fuzzy form. The model will mathematically combine cyberattacks like Denial of Service (DoS), Replay Attacks, and False Data Injection (FDI) to mimic real-world communication problems. Every system model will be used in MATLAB/Simulink for implementation and validation.

b) Design of Fuzzy Observer

A fuzzy observer will be developed in the following phase to estimate unidentified system states and use residual evaluation to find anomalies. To generate remaining data that point to possible attacks, the observer will compare expected outputs with actual sensor feedback. To

distinguish between normal and abnormal operation, a residual threshold will be set. Linear Matrix Inequalities (LMIs) will be used to determine observer gain matrices in order to guarantee estimation stability and quick detection response.

c) Development of Security-Weighted Fuzzy Inference

The next phase will introduce its main novelty, a security-weighted fuzzy inference mechanism. Trust coefficients will be computed using the fuzzy observer's residual magnitudes.

These coefficients can mitigate the effects of corrupted inputs by dynamically modifying the activation strength of the fuzzy rules. The modified membership functions will make the controller self-defensive, enabling real-time flexibility without the requirement for neural network tuning or data-driven learning.

d) Dual-Surface Lyapunov–LMI Controller Design

We will define two Lyapunov functions: one for regular operation and another for attack scenarios.

The controller gain matrices that meet stability restrictions for both regimes will be calculated using LMI-based optimization. This ensures that the STSFC may switch between operating modes smoothly and without experiencing instability or performance degradation.

e) Simulation and Experimental Validation

The proposed controller will be implemented on the **inverted pendulum benchmark** to evaluate its performance under simulated attack scenarios. Attack conditions will include biased sensor data (FDI), intermittent control signal loss (DoS), and delayed or replayed packets (DDoS/Replay).

Performance metrics will include:

- stabilization accuracy,
- control effort,
- Lyapunov decay rate, and
- resilience recovery time after an attack.

Comparative analysis will be performed against baseline TS fuzzy and adaptive fuzzy controllers to demonstrate the STSFC's superiority

5. Research Steps

A. Research Foundation and Framework Development for a Secure Ts-Fuzzy Controller, Aug. 2026~July. 2027

A1. First one and two months:

A thorough **literature study** of current TS fuzzy control schemes, secure and resilient control techniques, and new developments in cyber-physical system (CPS) security will be the main focus of Months 1-2. This stage will guarantee a solid theoretical basis, identify gaps in knowledge, and establish the exact parameters of the research and its assumptions. The classification of cyberattacks, particularly DoS, FDI, and replay attacks, and their mathematical models in control theory will get a special focus.

A2. Third and Fourth month:

In Months 3 and 4, the **development of the mathematical model of the system and the dynamics** of the attack will be the main focus. The nonlinear plant will be modeled using the Takagi–Sugeno fuzzy modeling approach, and analytical models for different kinds of cyberattacks will be created, including time-based stay functions for DoS, additional disruption formulas for FDI, and time-delay mechanisms for replay attacks. This step will produce an

integrated mathematical framework that integrates attack dynamics into the fuzzy control model.

A3. Fifth, Sixth and Seventh month:

During 5 to 7 months will be **spent to the controller design and stability analysis stage**. Furthermore, the secure TS fuzzy control rule is formally derived using a double-surface Lyapunov architecture and security-weighted membership functions. A Lyapunov-Krasovskii functional will be built, and appropriate Linear Matrix Inequality (LMI) conditions guaranteeing asymptotic stability and performance will be constructed. MATLAB and the SeDuMi simulators will be used to find workable solutions for controller gains and observer gains. The mathematical results will be used to determine the overall idea of secure fuzzy stabilization over cyberattack.

A4. Eighth and Ninth month:

The primary focus of months 8 and 9 will be **MATLAB R2024b simulation** and implementation. Once benchmark nonlinear systems such as the DC microgrid, Lorenz attractor, and inverted pendulum are created, the proposed controller will be tested under several attack scenarios. Performance metrics such Lyapunov energy decay rate, steady-state error, control effort, and attack rejection ratio (ARR) will be evaluated. Comparative analyses of traditional fuzzy controllers will also be carried out in order to determine the improvement achieved by the proposed secure design.

A5. Tenth and Eleventh month:

In Months 10 and 11 **performance evaluation, sensitivity testing, and result validation** shall be focused. We will check the controller's robustness against different attack intensities, packet loss rates, and replay delays. Security and performance will be balanced by modifying the security-weighted membership sensitivities parameter γ , and system stability and resilience can be maximized via parameter adjustments.

A6. Twelfth month:

Finally, in Month 12, the **documentation and presentation** step will take place. All findings, theoretical derivations, and simulation data would be included in a comprehensive final report and research article written for submission to renowned journals such as IEEE Transactions on Fuzzy Systems or IEEE Transactions on Cybernetics. Additionally, the public will have access to an open-source toolkit that includes the MATLAB implementation and LMI design scripts, which will ease replication and promote further research in academia.

6. (Schedule):

A Secure TS Fuzzy Control Framework for Nonlinear Systems Under Cyberattacks								
執行年月 工作項目	115年 8月	115年 9月	115年 10月	115年 12月	116年 1月	116年 4月	116年 6月	116年 7月
	Literature Collection							
Systematic Review and Synthesis								
Formulation and System Modeling								
Skill Development and Implementation Training								
Design of Fuzzy Observer								
Development of Security-Weighted Fuzzy Inference								
Lyapunov–LMI Controller Design								
Evaluation and Sensitivity Analysis								
Manuscript Preparation and Submission								
預定進度	10%	20%	30%	40%	50%	60%	90%	100%

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